

1. Configure Hostnames

```
Memory usage: 1%                               IPv4 address for ens33: 192.168.0.3
Swap usage:   0%

335 updates can be applied immediately.
239 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu May  7 09:45:51 2026 from 49.50.73.97
root@WorkerNode1:~# hostnamectl set-hostname worker1
root@WorkerNode1:~# apt update -y
apt install -y apt-transport-https ca-certificates curl gpg containerd
```

2. Basic Package Setup on All VMs

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu May  7 09:45:51 2026 from 49.50.73.97
root@WorkerNode1:~# hostnamectl set-hostname worker1
root@WorkerNode1:~# apt update -y
apt install -y apt-transport-https ca-certificates curl gpg containerd
```

```
Last login: Thu May  7 09:45:51 2026 from 49.50.73.97
root@MasterNode:~# hostnamectl set-hostname master
root@MasterNode:~# apt update -y
apt install -y apt-transport-https ca-certificates curl gpg containerd
```

3. Disable Swap and Enable Required Kernel Modules on All VMs

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@WorkerNode1:~# swapoff -a
sed -i '/ swap / s/^/#/' /etc/fstab
root@WorkerNode1:~# cat <<EOF | tee /etc/modules-load.d/k8s.conf
overlay
br_netfilter
EOF
root@WorkerNode1:~# modprobe overlay
modprobe br_netfilter
root@WorkerNode1:~# cat <<EOF | tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables=1
```

4. Configure Kernel Networking Parameters on all VMs

```
root@MasterNode:~# cat <<EOF | tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables=1
net.bridge.bridge-nf-call-ip6tables=1
net.ipv4.ip_forward=1
EOF
net.bridge.bridge-nf-call-iptables=1
net.bridge.bridge-nf-call-ip6tables=1
net.ipv4.ip_forward=1
root@MasterNode:~# sysctl --system
* Applying /etc/sysctl.d/10-console-messages.conf ...
kernel.printk = 4 4 1 7
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
kernel.kptr_restrict = 1
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
kernel.sysrq = 176
```

5. Configure containerd on All VMs

```
* Applying /etc/sysctl.conf ...
root@WorkerNode1:~# mkdir -p /etc/containerd
containerd config default | tee /etc/containerd/config.toml
sed -i 's/SystemdCgroup = false/SystemdCgroup = true/' /etc/containerd/config.toml
systemctl restart containerd
systemctl enable containerd
version = 3
root = '/var/lib/containerd'
state = '/run/containerd'
temp = ''
disabled_plugins = []
```

```
root@worker2: ~
Command 'crictl' not found, did you mean:
  command 'critcl' from deb critcl (3.1.18.1+dfsg-3)
Try: apt install <deb name>
root@worker2:~# systemctl status containerd --no-pager
● containerd.service - containerd container runtime
   Loaded: loaded (/lib/systemd/system/containerd.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-06-26 07:12:39 UTC; 26min ago
     Docs: https://containerd.io
   Main PID: 7738 (containerd)
      Tasks: 43
     Memory: 1.1G
        CPU: 30.341s
    CGroup: /system.slice/containerd.service
            └─ 7738 /usr/bin/containerd
               8709 /usr/bin/containerd-shim-runc-v2 -namespace k8s.io -id f2l...
               8727 /usr/bin/containerd-shim-runc-v2 -namespace k8s.io -id bf0...
              14506 /usr/bin/containerd-shim-runc-v2 -namespace k8s.io -id b5c...
```

6. Install Kubernetes Packages on All VMs

```
root@worker1:~# mkdir -p -m 755 /etc/apt/keyrings
curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.36/deb/Release.key | gpg --dearmor
  or -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
```

Install kubeadm, kubelet, and kubectl:

```
root@MasterNode:~# apt update -y
apt install -y kubelet kubeadm kubectl
apt-mark hold kubelet kubeadm kubectl
systemctl enable kubelet
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 https://prod-cdn.packages.k8s.io/repositories/iscv:/kubernetes:/core:/stable:/v1.36/deb InRelease [1,227 B]
Get:3 https://prod-cdn.packages.k8s.io/repositories/iscv:/kubernetes:/core:/stable:/v1.36/deb Packages [5,666 B]
Hit:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:6 http://archive.ubuntu.com/ubuntu jammy-security InRelease
Fetched 6,893 B in 1s (5,668 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

7. Initialize Kubernetes Master Node

Run only on the master node.

```
root@master:~# kubectl cluster-info
Kubernetes control plane is running at https://192.168.0.2:6443
CoreDNS is running at https://192.168.0.2:6443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
root@master:~#

root@master:~# kubectl get componentstatuses
Warning: v1 ComponentStatus is deprecated in v1.19+
NAME                 STATUS    MESSAGE             ERROR
controller-manager   Healthy   ok
scheduler             Healthy   ok
etcd-0                Healthy   ok
```

Install Calico network plugin:

```

root@master:~# kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
calico-kube-controllers-c66555cf8-jsj67  1/1     Running   0           3d3h
calico-node-4m6nz                      1/1     Running   0           3d2h
calico-node-jv7n7                      1/1     Running   0           3d2h
calico-node-vdctk                      1/1     Running   0           3d2h
calico-node-wfg6m                      1/1     Running   0           3d3h
coredns-589f44dc88-bf41t              1/1     Running   0           3d3h
coredns-589f44dc88-lchfx              1/1     Running   0           3d3h
etcd-master                           1/1     Running   0           3d3h
kube-apiserver-master                 1/1     Running   0           3d3h
kube-controller-manager-master        1/1     Running   0           3d3h
kube-proxy-2v564                      1/1     Running   0           3d2h
kube-proxy-9d4qv                     1/1     Running   0           3d3h
kube-proxy-pckgv                     1/1     Running   0           3d2h
kube-proxy-vph62                     1/1     Running   0           3d2h
kube-scheduler-master                 1/1     Running   0           3d3h

```

8. Join Worker Nodes and Verify Cluster

```

root@MasterNode:~# kubectl get nodes
NAME        STATUS   ROLES    AGE   VERSION
master      Ready    control-plane  33m   v1.36.2
worker1     Ready    <none>    9m13s v1.30.14
worker2     Ready    <none>    12m   v1.36.2
worker3     Ready    <none>    81s   v1.36.2

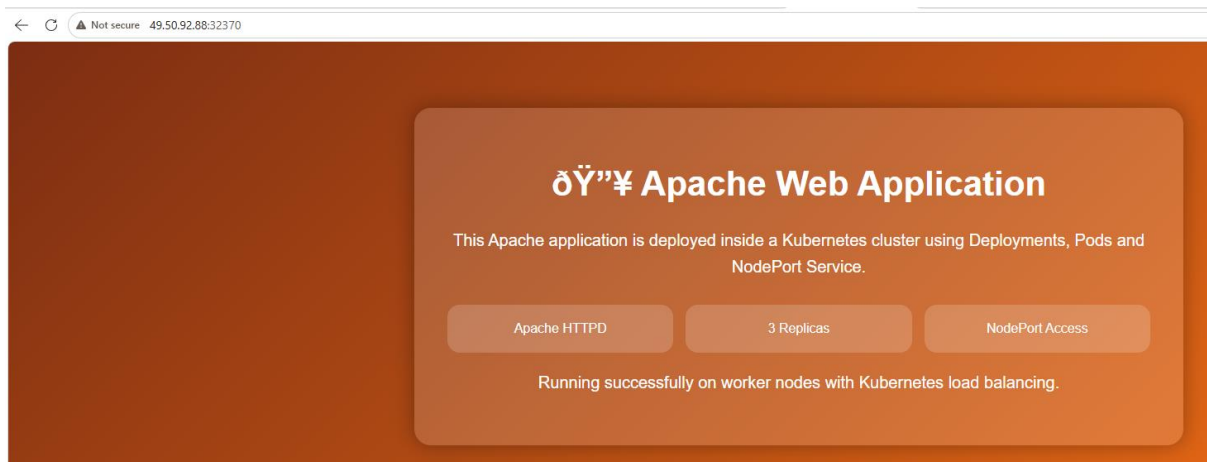
```

9. Deploy Apache Web Application

```

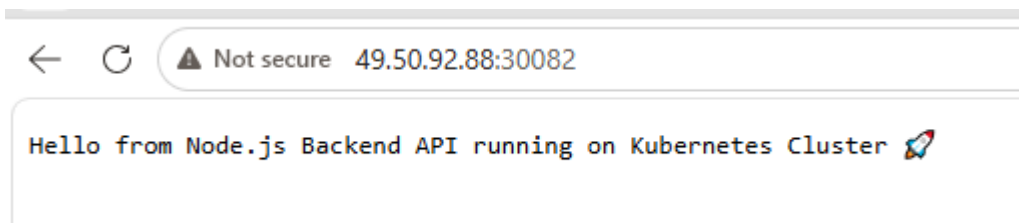
root@MasterNode:~# cat <<'EOF' > apache-website.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: apache-website
data:
  index.html: |
    <!DOCTYPE html>
    <html>
    <head>
      <title>Apache App on Kubernetes</title>
      <style>
        body {
          margin: 0;
          font-family: Arial, sans-serif;
          background: linear-gradient(135deg, #7c2d12, #f97316);
          color: white;
          text-align: center;
        }
        .container {
          margin: 80px auto;
          max-width: 850px;
          padding: 40px;
          border-radius: 20px;
        }
    EOF </html>>
root@MasterNode:~# kubectl apply -f apache-website.yaml
configmap/apache-website created

```



10. Deploy Node.js Backend API

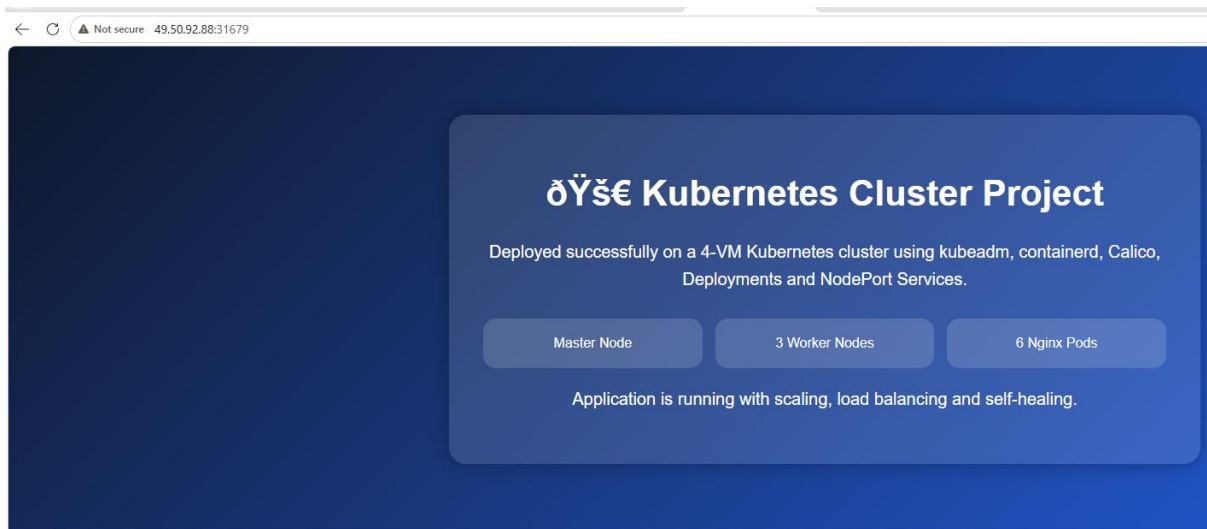
```
root@MasterNode:~# cat <<'EOF' > nodejs-api.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nodejs-api
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nodejs-api
  template:
    metadata:
      labels:
        app: nodejs-api
    spec:
      containers:
      - name: nodejs-api
        image: hashicorp/http-echo
        args:
          - "-text=Hello from Node.js Backend API running on Kubernetes Cluster"
        ports:
          - containerPort: 5678
---
EOF nodePort: 30082service
root@MasterNode:~# kubectl apply -f nodejs-api.yaml
deployment.apps/nodejs-api created
service/nodejs-api-service created
```



11. Deploy Nginx Web Application

```
root@MasterNode:~# kubectl create deployment nginx-app --image=nginx --replicas=3
deployment.apps/nginx-app created
root@MasterNode:~# kubectl expose deployment nginx-app --type=NodePort --port=80
service/nginx-app exposed
root@MasterNode:~# kubectl get pods -o wide
```

```
root@MasterNode:~# cat <<'EOF' > nginx-website.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nginx-website
data:
  index.html: |
    <!DOCTYPE html>
    <html>
    <head>
      <title>Kubernetes Cluster Project</title>
      <style>
        body {
          margin: 0;
          font-family: Arial, sans-serif;
          background: linear-gradient(135deg, #0f172a, #2563eb);
          color: white;
          text-align: center;
        }
        .box {
          margin: 80px auto;
          max-width: 800px;
          padding: 40px;
          border-radius: 20px;
EOF
application is running with scaling, load balancing and self-healing
root@MasterNode:~# kubectl apply -f nginx-website.yaml
configmap/nginx-website created
root@MasterNode:~# kubectl patch deployment nginx-app --type='json' -p='[
  {
    "op": "add",
    "path": "/spec/template/spec/volumes",
    "value": [
      {
        "name": "nginx-website-volume",
        "configMap": {
          "name": "nginx-website"
        }
      }
    ]
  },
  {
    "op": "add",
    "path": "/spec/template/spec/containers/0/volumeMounts",
    "value": [
      {
        "name": "nginx-website-volume",
        "mountPath": "/usr/share/nginx/html"
      }
    ]
  }
]'
```



12. Deploy MySQL Database

```
root@master: ~  
| kubernetesdb |  
| mysql        |  
| performance_schema |  
| sys          |  
+-----+  
5 rows in set (0.00 sec)  
  
mysql> USE kubernetesdb;  
Database changed  
mysql> CREATE TABLE students (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR  
(50), project VARCHAR(100));  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> INSERT INTO students (name, project) VALUES ('Chand', 'Kubernetes Clus  
ter Project');  
Query OK, 1 row affected (0.02 sec)  
  
mysql> SELECT * FROM students;  
+----+-----+-----+  
| id | name  | project |  
+----+-----+-----+  
| 1  | Chand | Kubernetes Cluster Project |  
+----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql> EXIT;
```


13. Deploy Redis Cache

```
root@master:~# cat <<'EOF' > redis.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: redis-cache
spec:
  replicas: 1
  selector:
    matchLabels:
      app: redis-cache
  template:
    metadata:
      labels:
        app: redis-cache
    spec:
      containers:
      - name: redis
        image: redis:7
        ports:
        - containerPort: 6379
---
apiVersion: v1
kind: Service
metadata:
  name: redis-service
spec:
  EOF targetPort: 6379
```

```
root@master:~# kubectl apply -f redis.yaml
kubectl get pods -o wide
kubectl get svc
```

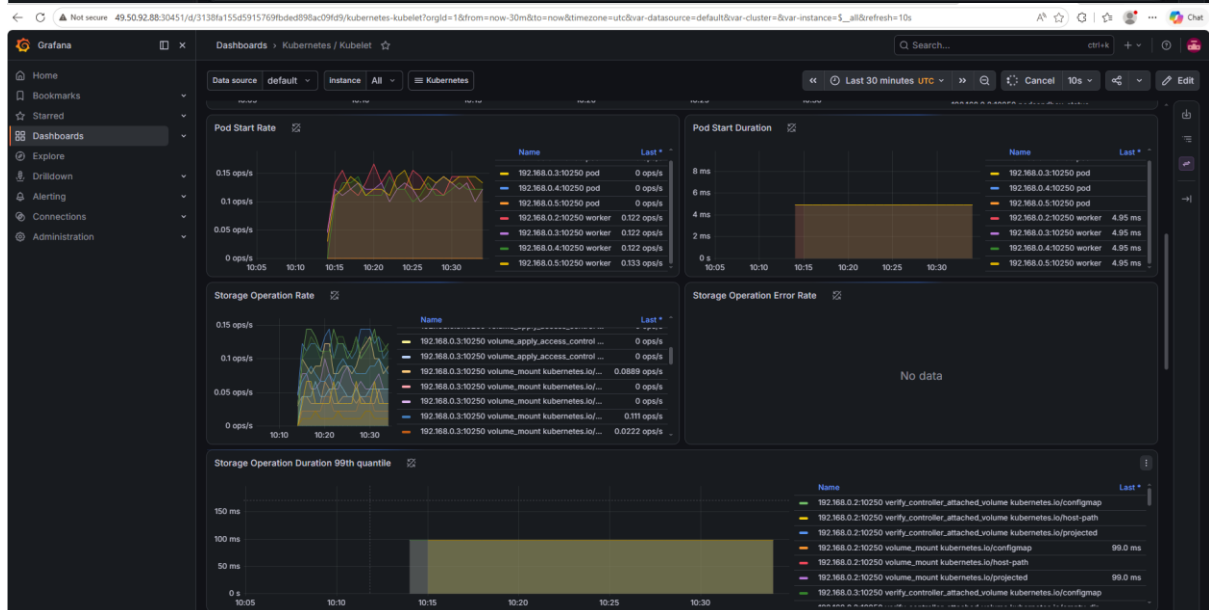
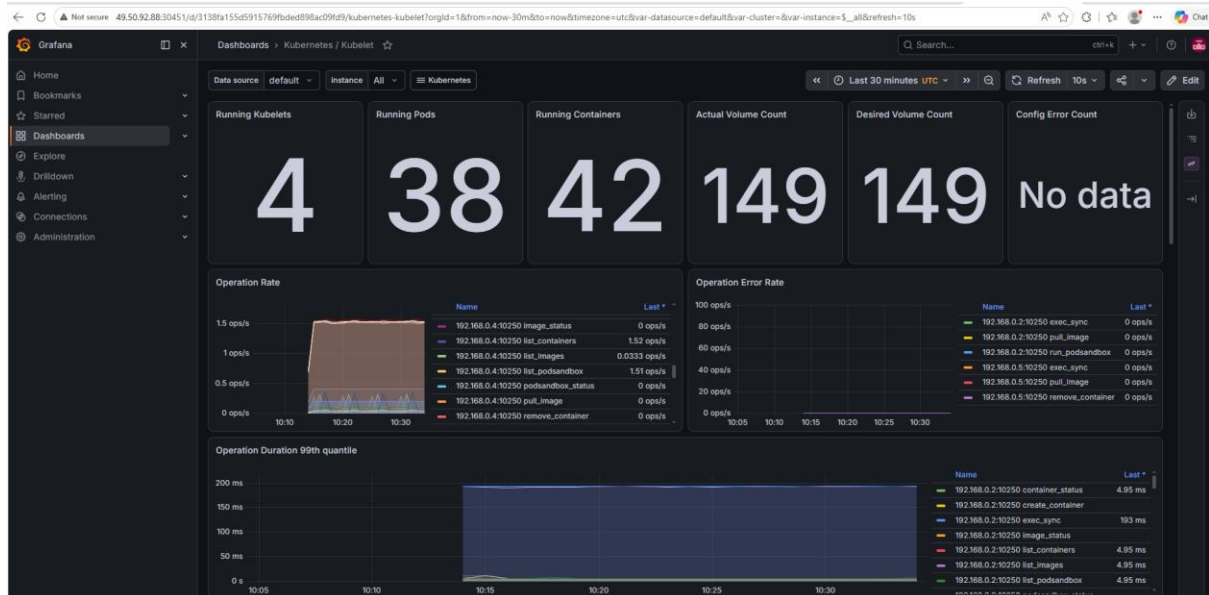
```
deployment.apps/redis-cache created
service/redis-service created
```

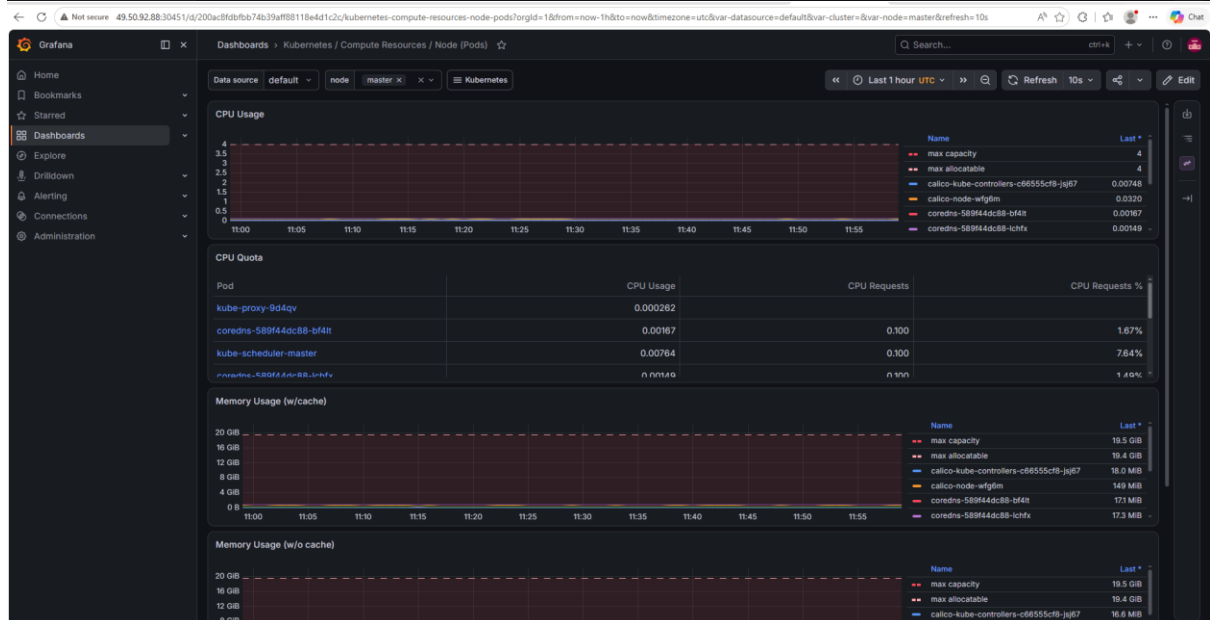
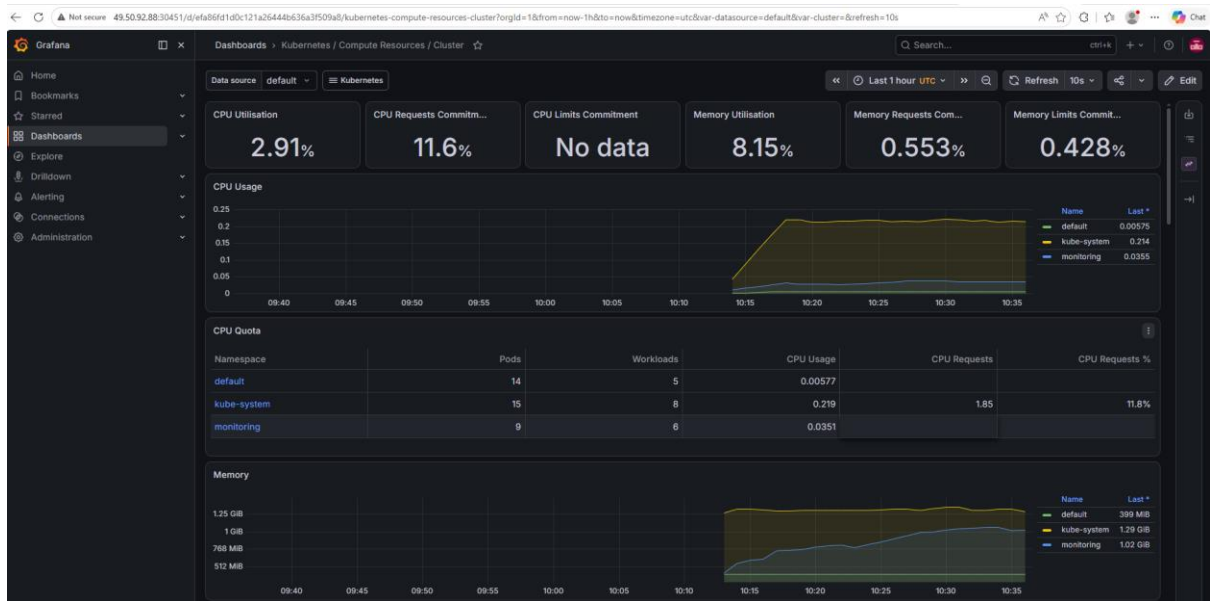
NAME	IP	NODE	READY	STATUS	RESTARTS	AGE
			NOMINATED	NODE	READINESS GATES	
apache-app-85f7d4dd9b-9vqhg	192.168.235.135	worker1	1/1	Running	0	95m
apache-app-85f7d4dd9b-s4xtt	192.168.182.7	worker3	<none>	<none>	0	95m
apache-app-85f7d4dd9b-vgqc6	192.168.189.71	worker2	<none>	<none>	0	95m
mysql-db-59655dc955-vrjpb	192.168.189.74	worker2	1/1	Running	0	5m32s
nginx-app-65b764f7bd-5jshj	192.168.182.6	worker3	<none>	<none>	0	97m
nginx-app-65b764f7bd-6hdm5	192.168.189.70	worker2	<none>	<none>	0	97m
nginx-app-65b764f7bd-9pvs7	192.168.189.69	worker2	1/1	Running	0	97m
nginx-app-65b764f7bd-nmjfr	192.168.182.5	worker3	<none>	<none>	0	97m
nginx-app-65b764f7bd-nzr5z	192.168.235.133	worker1	1/1	Running	0	97m
nginx-app-65b764f7bd-qprfh	192.168.235.134	worker1	<none>	<none>	0	97m
nodejs-api-75997c84bc-9gb86	192.168.182.8	worker3	1/1	Running	0	89m
nodejs-api-75997c84bc-hh2lq	192.168.189.72	worker2	<none>	<none>	0	89m
nodejs-api-75997c84bc-whpj8	192.168.235.136	worker1	1/1	Running	0	89m
redis-cache-5b8949d899-dbdcj	<none>	worker1	0/1	ContainerCreating	0	0s

14. Create Monitoring Namespace & Install Helm

```
Update Complete. ☐Happy Helming!☐
root@master:~# helm install monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring
NAME: monitoring
LAST DEPLOYED: Fri Jun 26 10:11:03 2026
NAMESPACE: monitoring
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
TEST SUITE: None
```

15. Install Prometheus and Grafana





16. Kubernetes Services and Grafana Authentication

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)
apac				
AGE				
apache-service	NodePort	10.107.156.106	<none>	80:32370/TCP
18m				
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP
104m				
mysql-service	ClusterIP	10.109.33.222	<none>	3306/TCP
7s				
nginx-app	NodePort	10.101.160.236	<none>	80:31679/TCP
68m				
nodejs-api-service	NodePort	10.102.250.37	<none>	5678:30082/TCP
7m3s				

17. Grafana Services and Monitoring Stack Deployment

```

NAME                                TYPE          CLUSTER-IP      EXTERN
AL-IP    PORT(S)                AGE
alertmanager-operated              ClusterIP      None             <none>
9093/TCP,9094/TCP,9094/UDP          2m20s
monitoring-grafana                  NodePort       10.105.17.59     <none>
80:30451/TCP                        2m41s
monitoring-kube-prometheus-alertmanager ClusterIP       10.101.29.6      <none>
9093/TCP,8080/TCP                   2m41s
monitoring-kube-prometheus-operator ClusterIP       10.111.56.27     <none>
443/TCP                             2m41s
monitoring-kube-prometheus-prometheus ClusterIP       10.109.178.154   <none>
9090/TCP,8080/TCP                   2m41s
monitoring-kube-state-metrics       ClusterIP       10.110.7.58      <none>
8080/TCP                             2m41s
monitoring-prometheus-node-exporter ClusterIP       10.104.48.207    <none>
9100/TCP                             2m41s
prometheus-operated                 ClusterIP      None             <none>
9090/TCP                             83s
root@master:~# kubectl get secret monitoring-grafana -n monitoring -o jsonpath="{.data.admin-password}" | base64 -d
echo
JSyXRXFwtZZ3e4x5QVGBRqgUPPVS3cwcqIC4lgV8
root@master:~#

```

18. Monitoring Namespace Pods Running Successfully

```

root@master:~# kubectl get pods -n monitoring
NAME                                READY   STATUS    RE
STARTS   AGE
alertmanager-monitoring-kube-prometheus-alertmanager-0  2/2     Running   0
98s
monitoring-grafana-58b6c8bdbb-wl4tx  3/3     Running   0
119s
monitoring-kube-prometheus-operator-69c59c45f9-c9944  1/1     Running   0
119s
monitoring-kube-state-metrics-559db896bf-m19rd  1/1     Running   0
119s
monitoring-prometheus-node-exporter-58gvp  1/1     Running   0
119s
monitoring-prometheus-node-exporter-bh9gd  1/1     Running   0
119s
monitoring-prometheus-node-exporter-lcrv1  1/1     Running   0
119s
monitoring-prometheus-node-exporter-sgg2m  1/1     Running   0
119s
prometheus-monitoring-kube-prometheus-prometheus-0  2/2     Running   0
41s

```

19. Multi-Application Kubernetes Cluster with Running Pods

```
root@master:~# kubectl get pods -o wide
```

NAME	NODE	NOMINATED	READY	STATUS	RESTARTS	AGE	IP
		NODE	READY	STATUS	RESTARTS	AGE	IP
apache-app-85f7d4dd9b-9vqhg	5.135	worker1	1/1	Running	0	95m	192.168.23
apache-app-85f7d4dd9b-s4xtt	2.7	worker3	1/1	Running	0	95m	192.168.18
apache-app-85f7d4dd9b-vgqc6	9.71	worker2	1/1	Running	0	95m	192.168.18
mysql-db-59655dc955-vrjpb	9.74	worker2	1/1	Running	0	6m3s	192.168.18
nginx-app-65b764f7bd-5jshj	2.6	worker3	1/1	Running	0	98m	192.168.18
nginx-app-65b764f7bd-6hdm5	9.70	worker2	1/1	Running	0	98m	192.168.18
nginx-app-65b764f7bd-9pvs7	9.69	worker2	1/1	Running	0	98m	192.168.18
nginx-app-65b764f7bd-nmjfr	2.5	worker3	1/1	Running	0	98m	192.168.18
nginx-app-65b764f7bd-nzr5z	5.133	worker1	1/1	Running	0	98m	192.168.23
nginx-app-65b764f7bd-qprfh	5.134	worker1	1/1	Running	0	98m	192.168.23
nodejs-api-75997c84bc-9gb86	2.8	worker3	1/1	Running	0	90m	192.168.18
nodejs-api-75997c84bc-hh2lq	9.72	worker2	1/1	Running	0	90m	192.168.18
nodejs-api-75997c84bc-whpj8	5.136	worker1	1/1	Running	0	90m	192.168.23
redis-cache-5b8949d899-dbdcj	5.137	worker1	1/1	Running	0	31s	192.168.23

20. Git Repo Status Before Initial Commit

```
root@master:~/kubernetes-multi-application-cluster# git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   apache-website.yaml
    new file:   mysql-clean.yaml
    new file:   nginx-website.yaml
    new file:   nodejs-api.yaml
    new file:   nodes-output.txt
    new file:   pods-output.txt
    new file:   redis.yaml
    new file:   services-output.txt

root@master:~/kubernetes-multi-application-cluster#
```

21. Successful Push of Project to Github Repo

```
root@master:~/kubernetes-multi-application-cluster# git push -u origin main
Username for 'https://github.com': Chand643
Password for 'https://Chand643@github.com':
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 4 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 3.63 KiB | 3.63 MiB/s, done.
Total 10 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Chand643/kubernetes-multi-application-cluster.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
root@master:~/kubernetes-multi-application-cluster#
```

